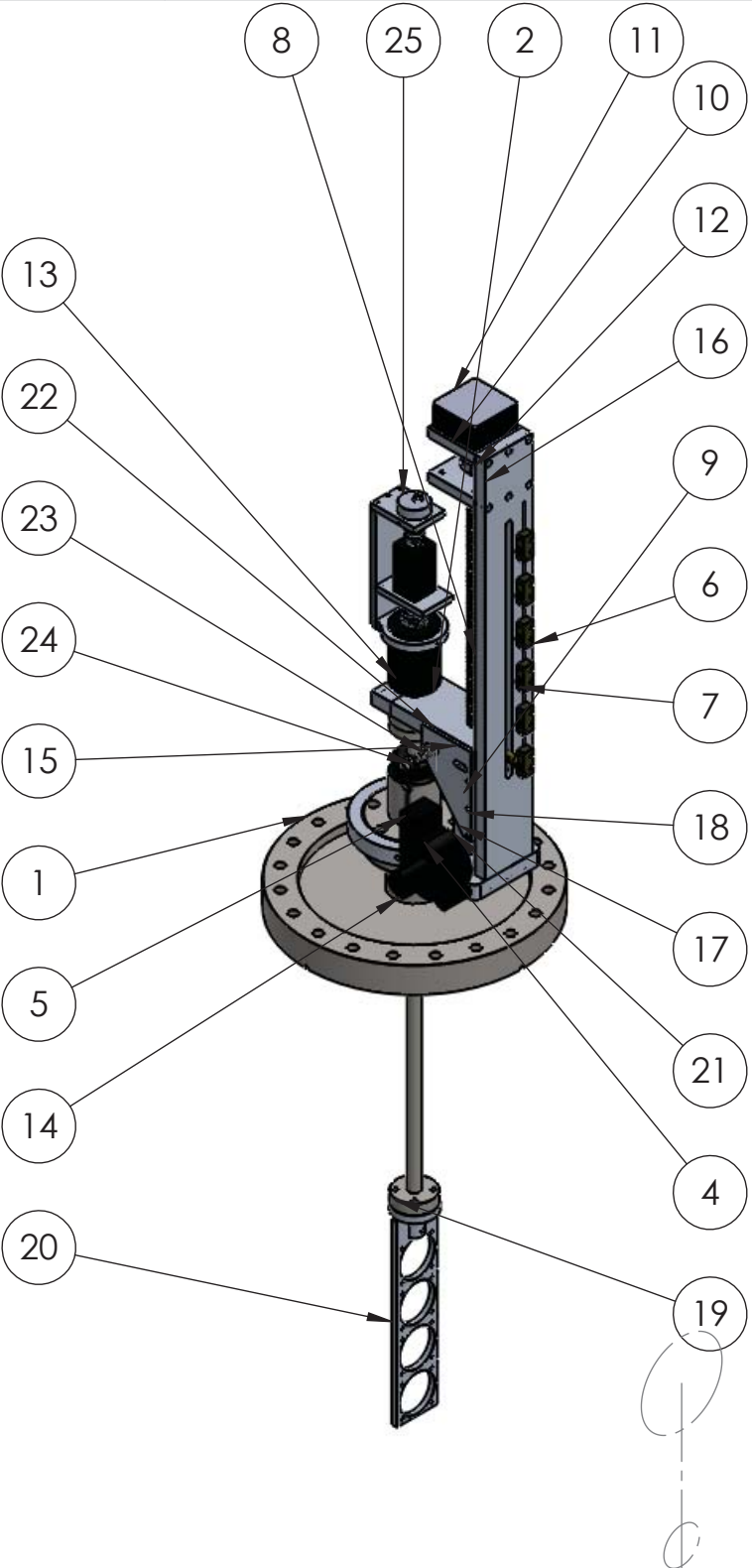


ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	8 conflat	Existing Part	1
2	Actuator	Existing Part	1
3	P102278	Conflat Flange (CF) Copper Gaskets CF Size 2-3/4 inch - 1 each	1
4	9088K15	Flange, Socket-Weld, 2-3/4" OD, for 1-1/2" Tube OD	1
5	Copper Gasket	Existing Part	1
6	Support_arm	Existing Part	1
7	microswitch	Existing Part	6
8	Ball Screw	Existing Part	1
9	ball_nut	Existing Part	1
10	Linear motor_mount	Existing Part	1
11	Linear motor	Existing Part	1
12	6208K432	Existing Part - Shaft Coupling	1
13	Slide_clamp	Existing Part	1
14	String pot	Existing Part	1
15	Clamp screw	Existing Part	1
16	Bearing_mount	Existing Part	1
17	Bearing_mount2	Existing Part	1
18	5908K12	Existing Part - Bearings	2
19	target_ladder1	Existing Part	1
20	target_ladder2	New Part	1
21	Conflat_clamp	Existing Part-Needs Modification	1
22	slide	Existing Part-Needs Modification	1
23	92185A227	New - 1.25" #10-32 Screw	1
24	90205A319	New - #10-32 NarrowNut	1
25	Rotational elements	Assembly- 9 parts	1
26	Lever_arm	Assembly - 9 parts	1
27	ASSY., CRYOSTAT_10-8-14		1



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		UNLESS OTHERWISE SPECIFIED:			NAME	DATE	TEMPLE UNIVERSITY	
		DIMENSIONS ARE IN INCHES	DRAWN		JPW	06/09/15	TITLE:	
		TOLERANCES:	CHECKED				Moller Target Arm	
		FRACTIONAL ±	ENG APPR.					
		ANGULAR: MACH ± BEND ±	MFG APPR.					
		TWO PLACE DECIMAL ±	Q.A.				SIZE DWG. NO. REV	
		THREE PLACE DECIMAL ±	COMMENTS:				B Target_arm 1	
		INTERPRET GEOMETRIC TOLERANCING PER:					SCALE: 1:20 WEIGHT: SHEET 1 OF 15	
		MATERIAL						
		Various						
		FINISH						
NEXT ASSY		USED ON						
APPLICATION		DO NOT SCALE DRAWING						

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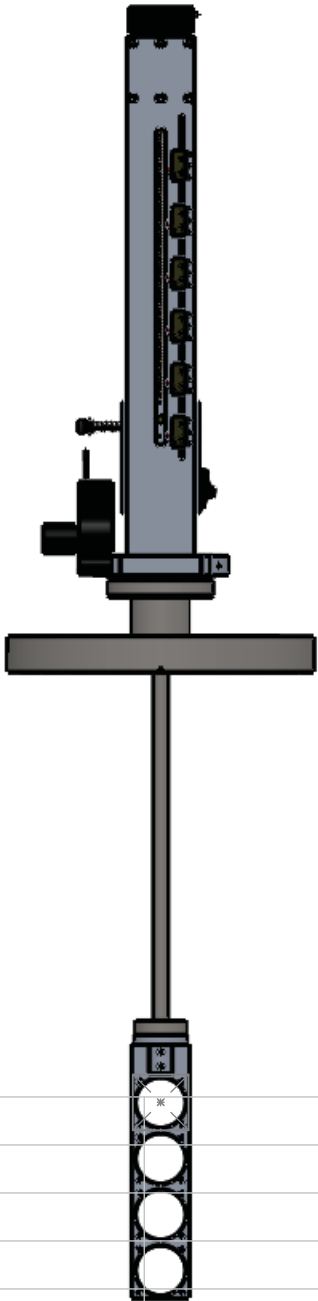
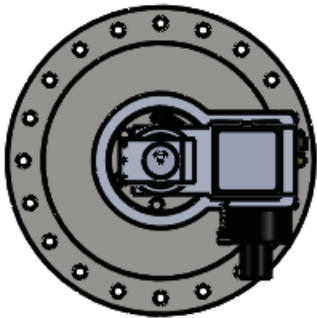
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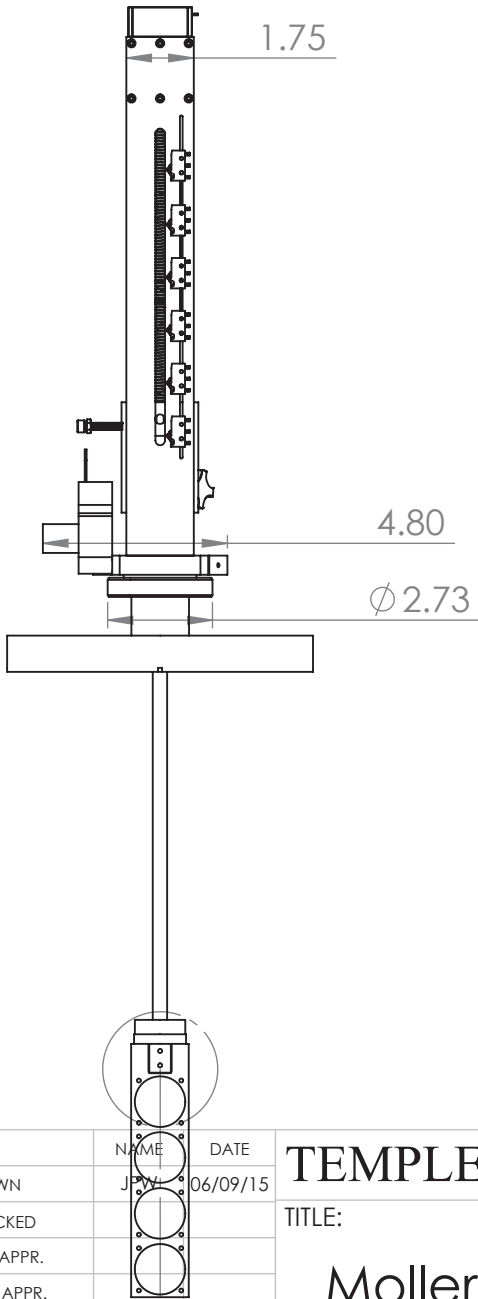
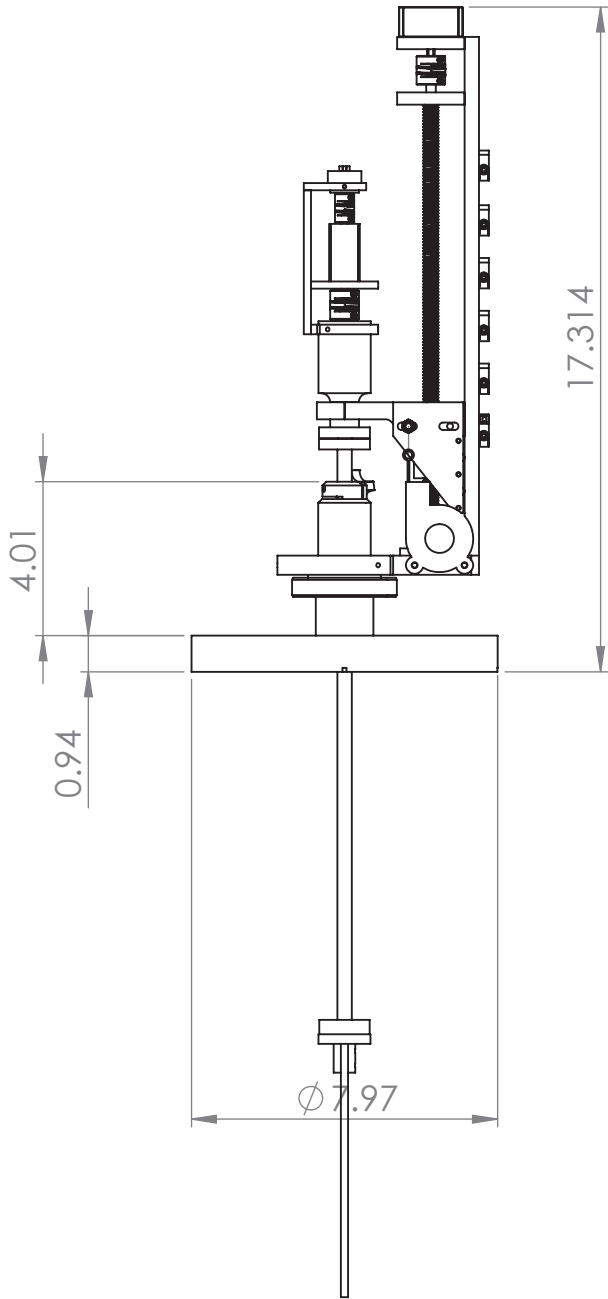
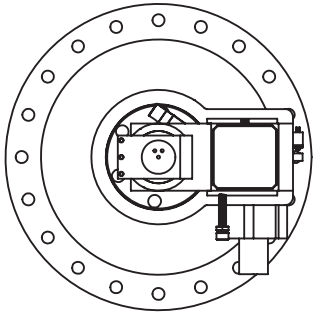
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				CHECKED			
				ENG APPR.			
		INTERPRET GEOMETRIC TOLERANCING PER:		MFG APPR.		TITLE:	
		MATERIAL		Q.A.			
				COMMENTS:		SIZE	DWG. NO.
NEXT ASSY	USED ON	FINISH				B	Target_arm
APPLICATION		DO NOT SCALE DRAWING				SCALE: 1:20	WEIGHT:
						SHEET 2 OF 15	

4

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2

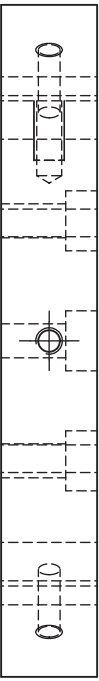
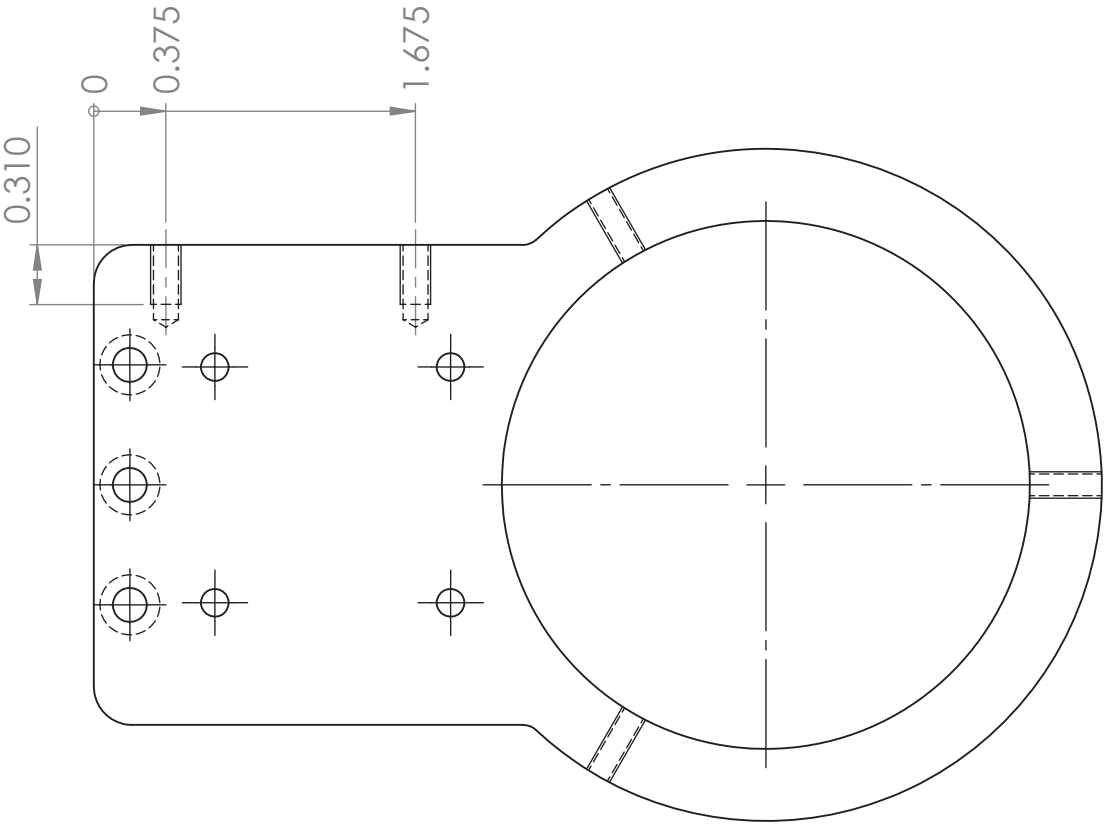
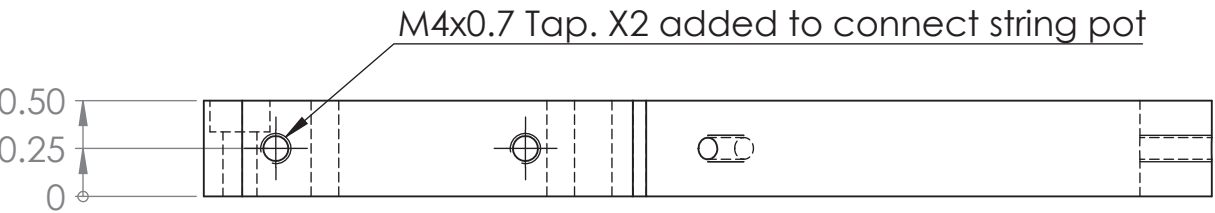
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		UNLESS OTHERWISE SPECIFIED:		NAME	DATE	TEMPLE UNIVERSITY	
		DIMENSIONS ARE IN INCHES	DRAWN	JPM	06/09/15	TITLE:	
		TOLERANCES:	CHECKED			Moller Target Arm	
		FRACTIONAL ±	ENG APPR.			SIZE DWG. NO. REV	
		ANGULAR: MACH ± BEND ±	MFG APPR.			B Target_arm 1	
		TWO PLACE DECIMAL ±	Q.A.			SCALE: 1:20 WEIGHT: SHEET 3 OF 15	
		THREE PLACE DECIMAL ±	COMMENTS:				
		INTERPRET GEOMETRIC TOLERANCING PER:					
		MATERIAL					
		Various					
		FINISH					
NEXT ASSY	USED ON						
APPLICATION		DO NOT SCALE DRAWING					

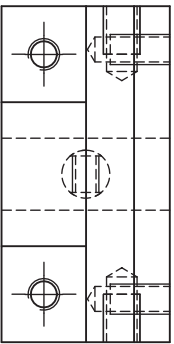
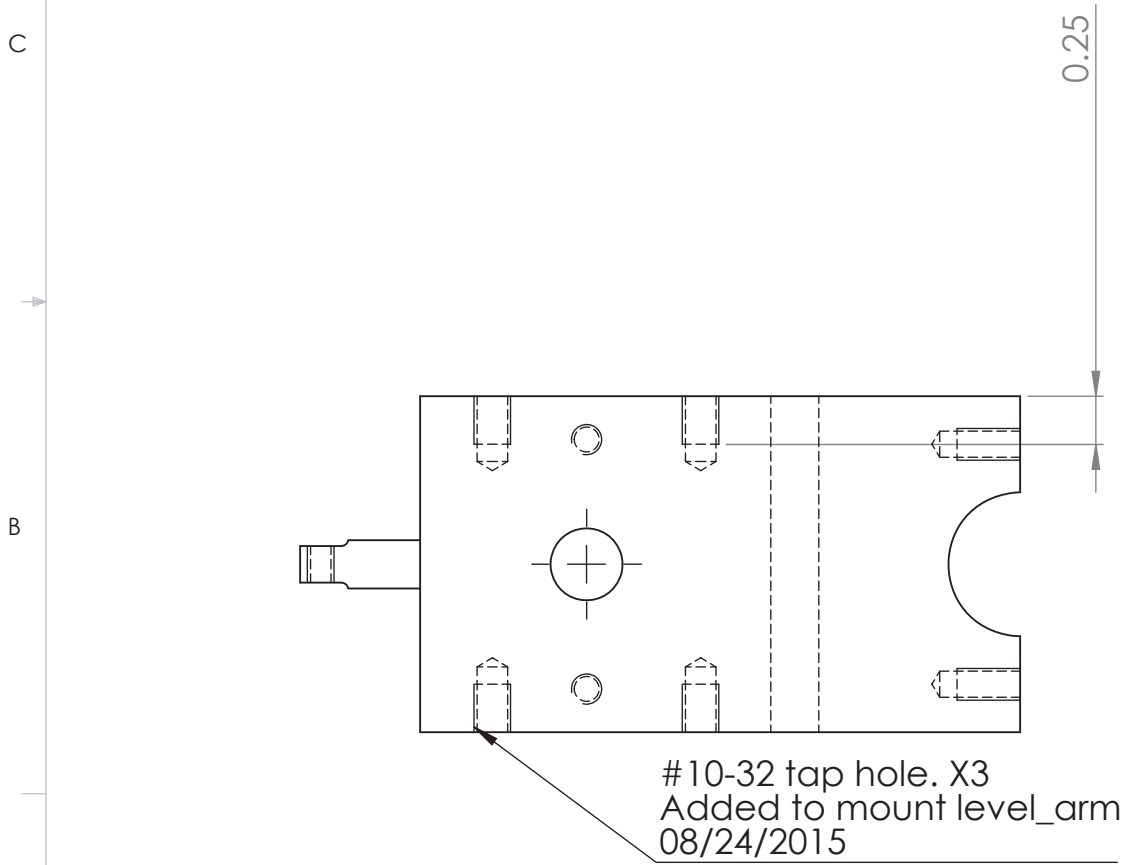
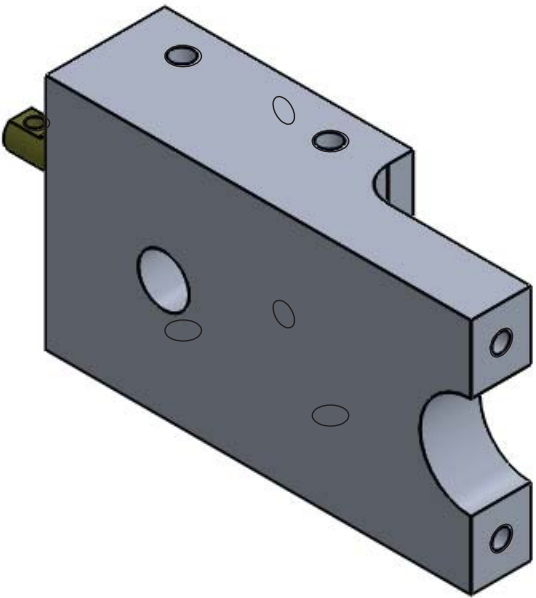
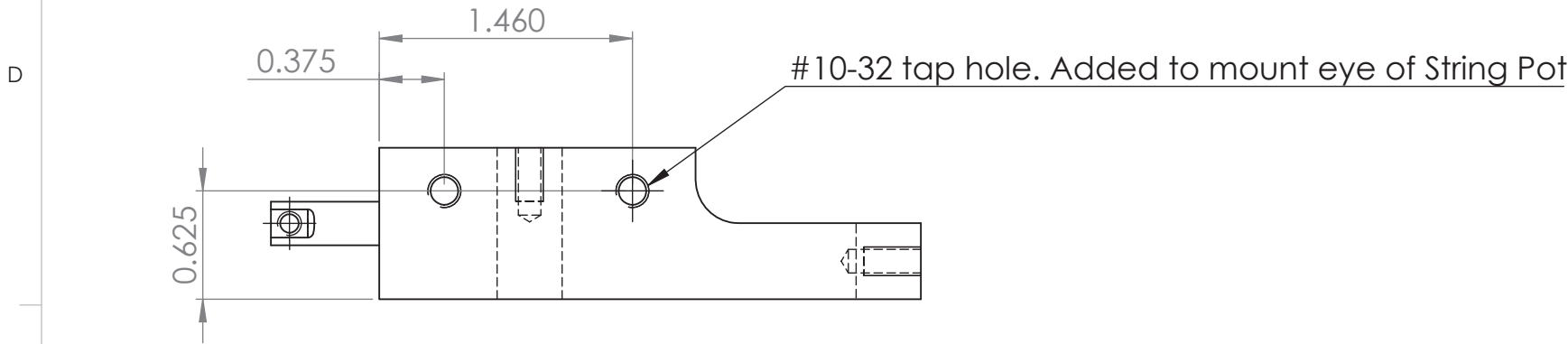
This is item #21 on the "Moller Target Arm" bill of materials. It is an existing part, which needs minor modifications.



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			DRAWN	JPW	06/09/15	TITLE:	
			CHECKED			Conflat Clamp	
			ENG APPR.			SIZE	DWG. NO.
			MFG APPR.			B	Target_arm
NEXT ASSY	USED ON		Q.A.			REV	1
APPLICATION		DO NOT SCALE DRAWING	COMMENTS:			SCALE: 1:20	WEIGHT:
						SHEET 4 OF 15	

Item #22 in "Moller Target Arm" bill of materials, this part needs modification to allow the eye of the string pot move with the actuator.

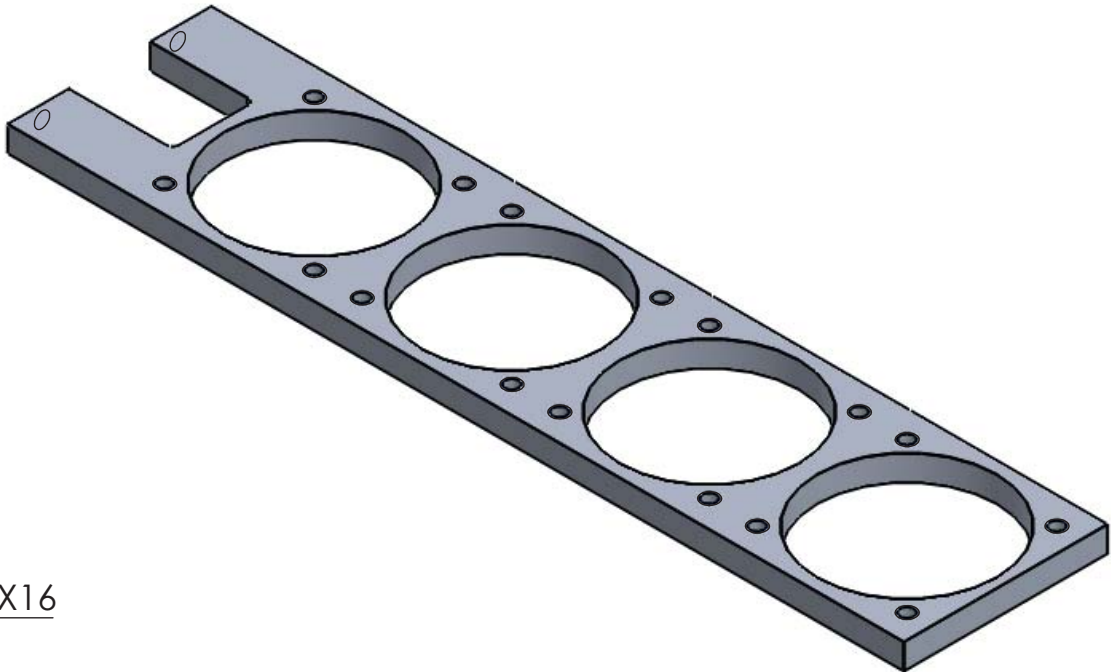
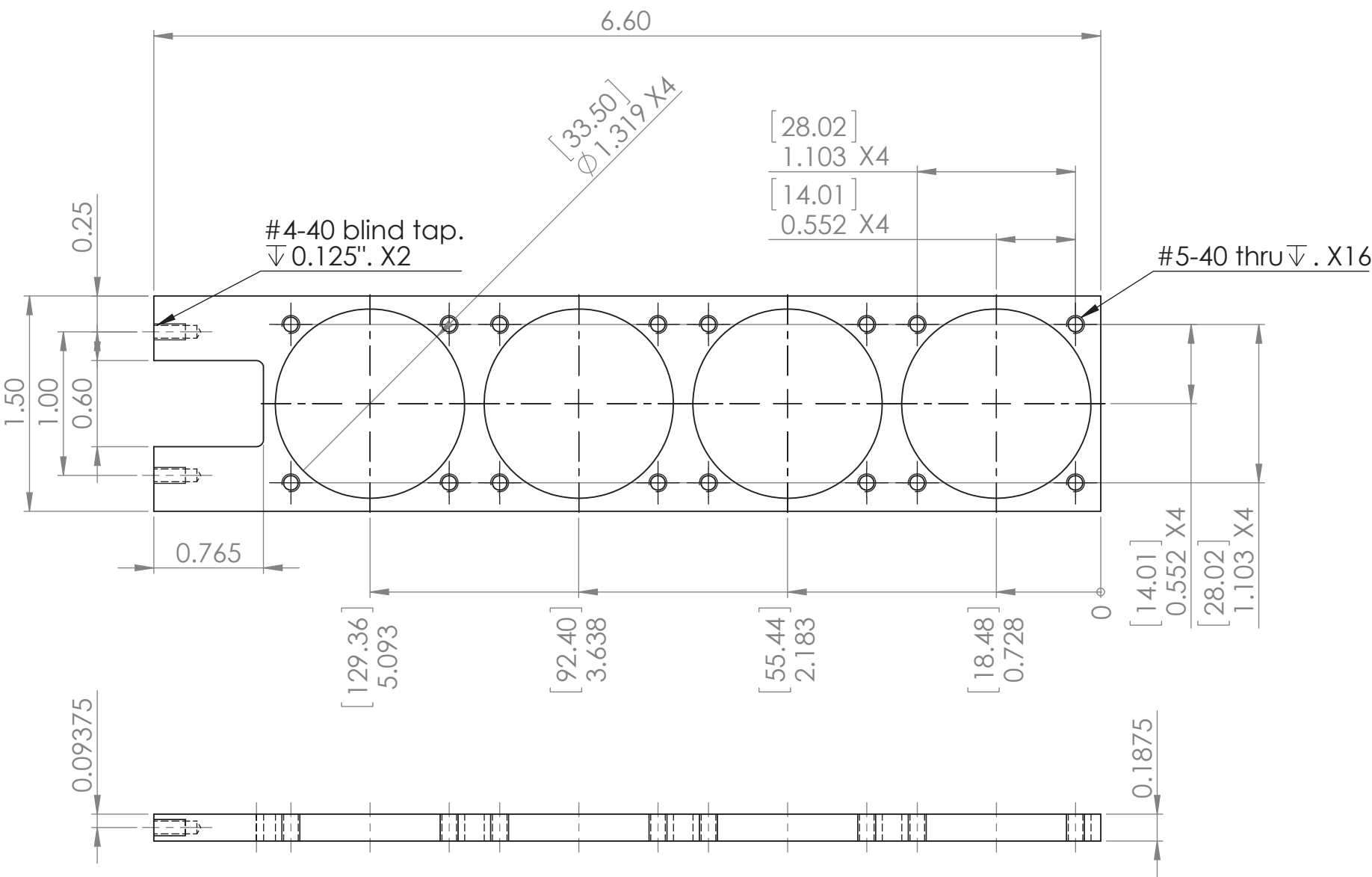


05/20/2015

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		UNLESS OTHERWISE SPECIFIED:		NAME	DATE	TEMPLE UNIVERSITY		
		DIMENSIONS ARE IN INCHES	DRAWN	JPW	06/09/15	TITLE: Slide		
		TOLERANCES:	CHECKED					
		FRACTIONAL ±	ENG APPR.					
		ANGULAR: MACH ± BEND ±	MFG APPR.					
		TWO PLACE DECIMAL ±	Q.A.			SIZE DWG. NO. REV B Target_arm 1		
		THREE PLACE DECIMAL ±	COMMENTS:					
		INTERPRET GEOMETRIC TOLERANCING PER:						
		MATERIAL						
		Aluminum						
NEXT ASSY	USED ON	FINISH						
APPLICATION		DO NOT SCALE DRAWING				SCALE: 1:20 WEIGHT: SHEET 5 OF 15		

Item #20 in "Moller Target Arm" bill of materials. Existing target ladder is to narrow for foil clamps.

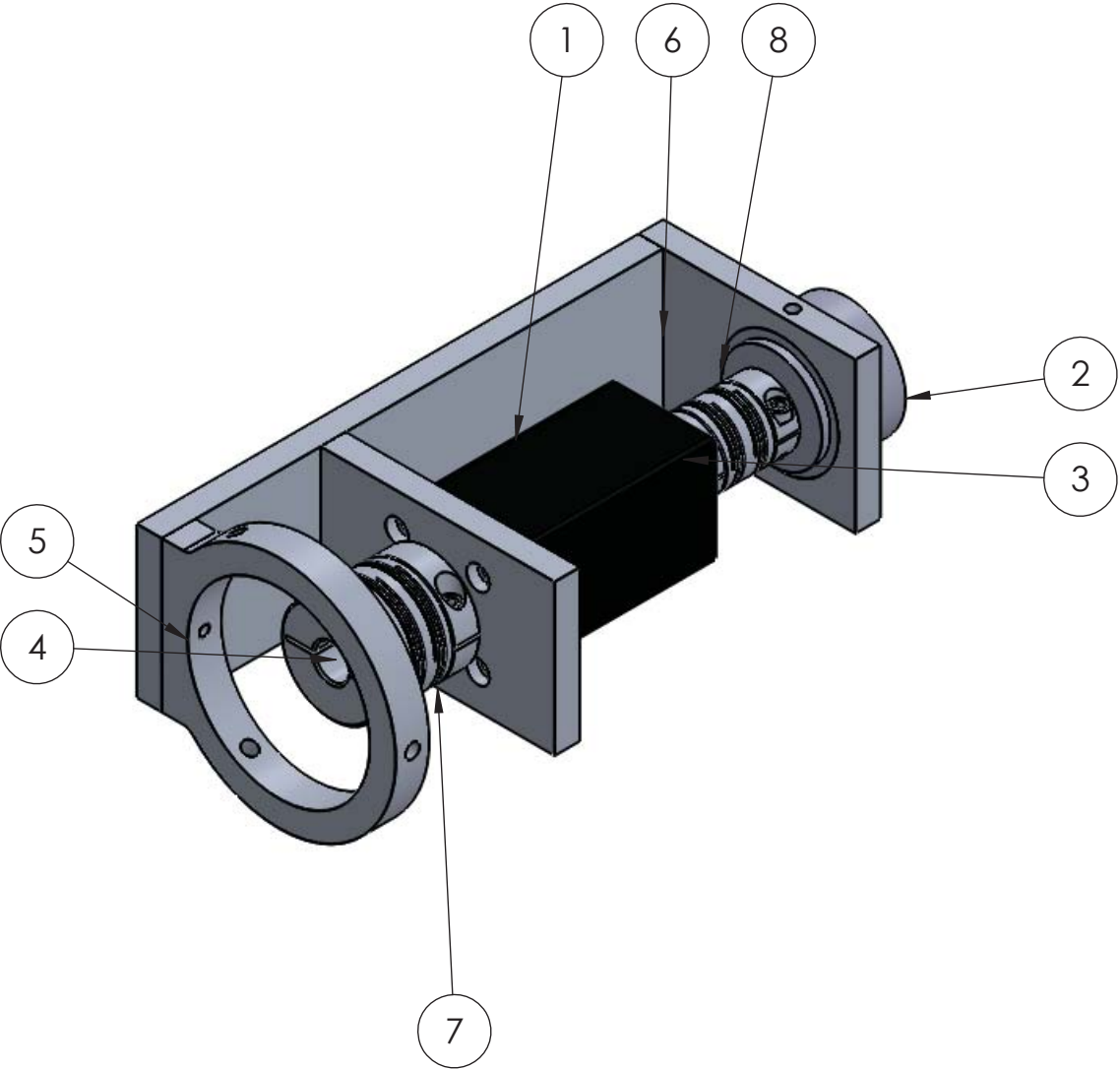


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		UNLESS OTHERWISE SPECIFIED:		NAME	DATE	TEMPLE UNIVERSITY		
		DIMENSIONS ARE IN INCHES	DRAWN	JPW	06/09/15			
		TOLERANCES:	CHECKED					
		FRACTIONAL ±	ENG APPR.					
		ANGULAR: MACH ±	BEND ±			TITLE: Target Ladder		
		TWO PLACE DECIMAL ±						
		THREE PLACE DECIMAL ±	MFG APPR.			SIZE DWG. NO. REV B Target_arm 1		
		INTERPRET GEOMETRIC TOLERANCING PER:	Q.A.					
		MATERIAL	COMMENTS:					
		Aluminum						
		FINISH				SCALE: 1:20 WEIGHT: SHEET 6 OF 15		
NEXT ASSY	USED ON							
APPLICATION		DO NOT SCALE DRAWING						

ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	Rotary motor	Existing Part	1
2	Rotary pot	Existing Part	1
3	6208K423	Stock Shaft Coupling	1
4	6208K431	Stock Shaft Coupling	1
5	Actuator_clamp	New Part	1
6	Rotary mount bar	New Part	1
7	Rotary motor mount	New Part	1
8	rotarty pot mount	New Part	1

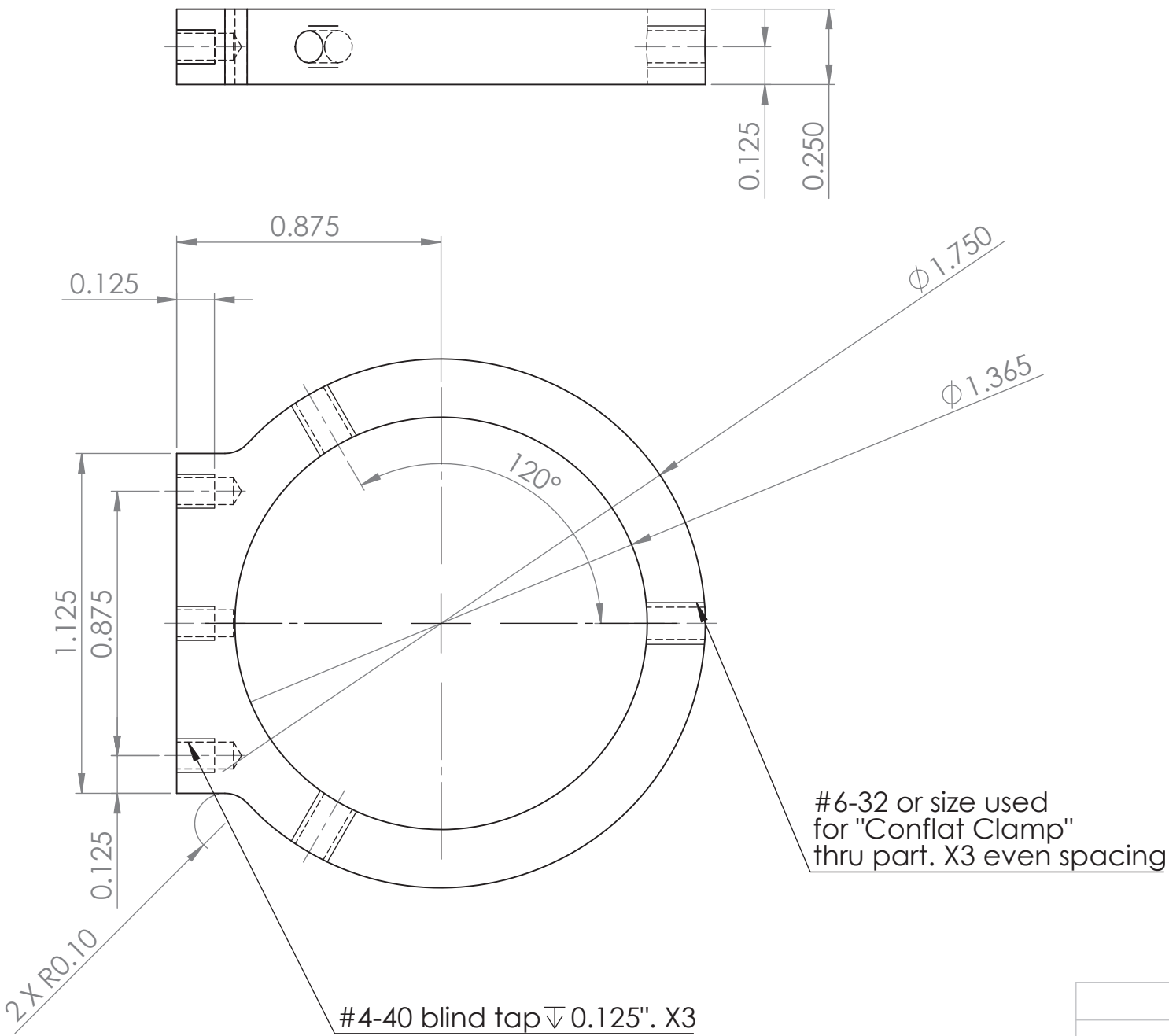
This is item #25 on the "Moller Target Arm" bill of materials. This is a new assembly.



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		UNLESS OTHERWISE SPECIFIED:		NAME	DATE	TEMPLE UNIVERSITY		
		DIMENSIONS ARE IN INCHES	DRAWN	JPW	06/09/15	TITLE: Rotational Elements Assembly		
		TOLERANCES:	CHECKED					
		FRACTIONAL ±	ENG APPR.					
		ANGULAR: MACH ± BEND ±	MFG APPR.					
		TWO PLACE DECIMAL ±				SIZE DWG. NO. REV B Target_arm 1		
		THREE PLACE DECIMAL ±						
		INTERPRET GEOMETRIC TOLERANCING PER:	Q.A.			SCALE: 1:20 WEIGHT: SHEET 7 OF 15		
		MATERIAL Various	COMMENTS:					
NEXT ASSY	USED ON	FINISH						
APPLICATION		DO NOT SCALE DRAWING						

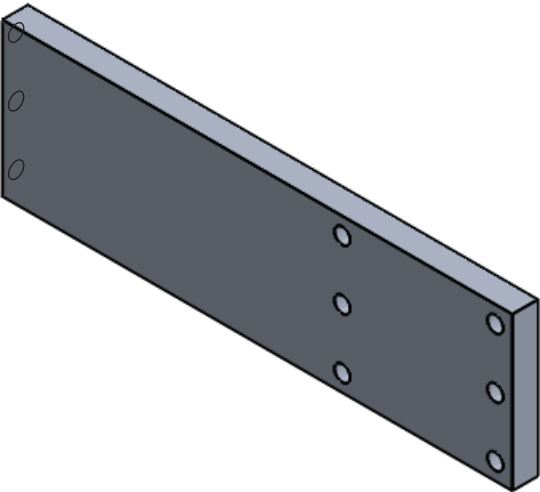
This is item #5 in bill of materials for "Rotational Elements Assembly". This is a new part which needs to be made.



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		UNLESS OTHERWISE SPECIFIED:		NAME	DATE	TEMPLE UNIVERSITY		
		DIMENSIONS ARE IN INCHES	DRAWN	JPW	06/09/15	TITLE:		
		TOLERANCES:	CHECKED			Actuator Clamp		
		FRACTIONAL ±	ENG APPR.			SIZE DWG. NO. REV		
		ANGULAR: MACH ± BEND ±	MFG APPR.			B Target_arm 1		
		TWO PLACE DECIMAL ±	Q.A.			SCALE: 1:20 WEIGHT: SHEET 8 OF 15		
		THREE PLACE DECIMAL ±	COMMENTS:					
		INTERPRET GEOMETRIC TOLERANCING PER:						
		MATERIAL						
		Aluminum						
NEXT ASSY	USED ON	FINISH						
APPLICATION		DO NOT SCALE DRAWING						

This is item #6 in bill of materials for "Rotational Elements Assembly". This is a new part which needs to be made.



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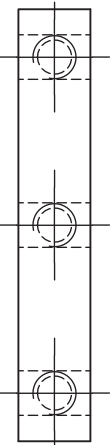
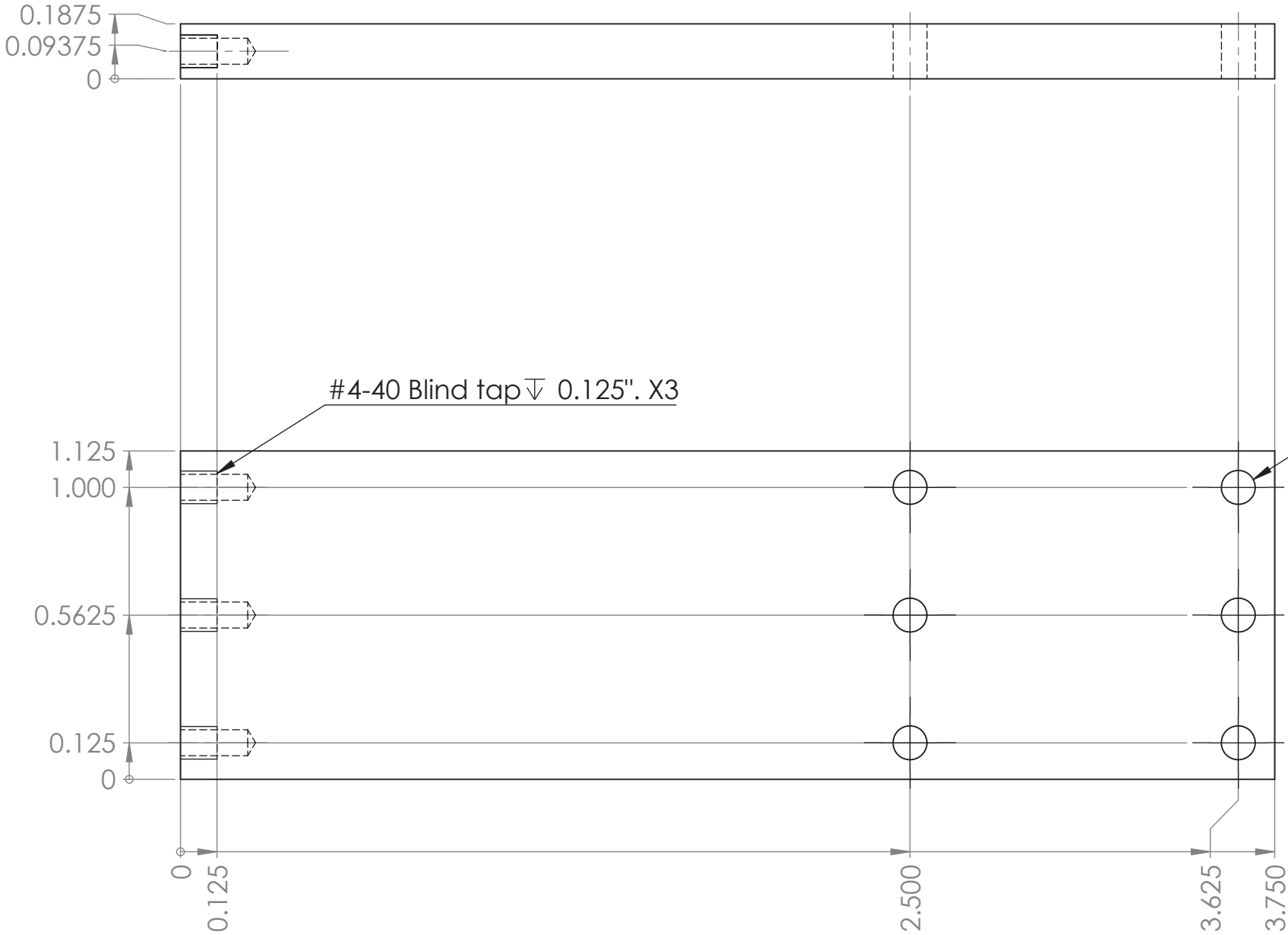
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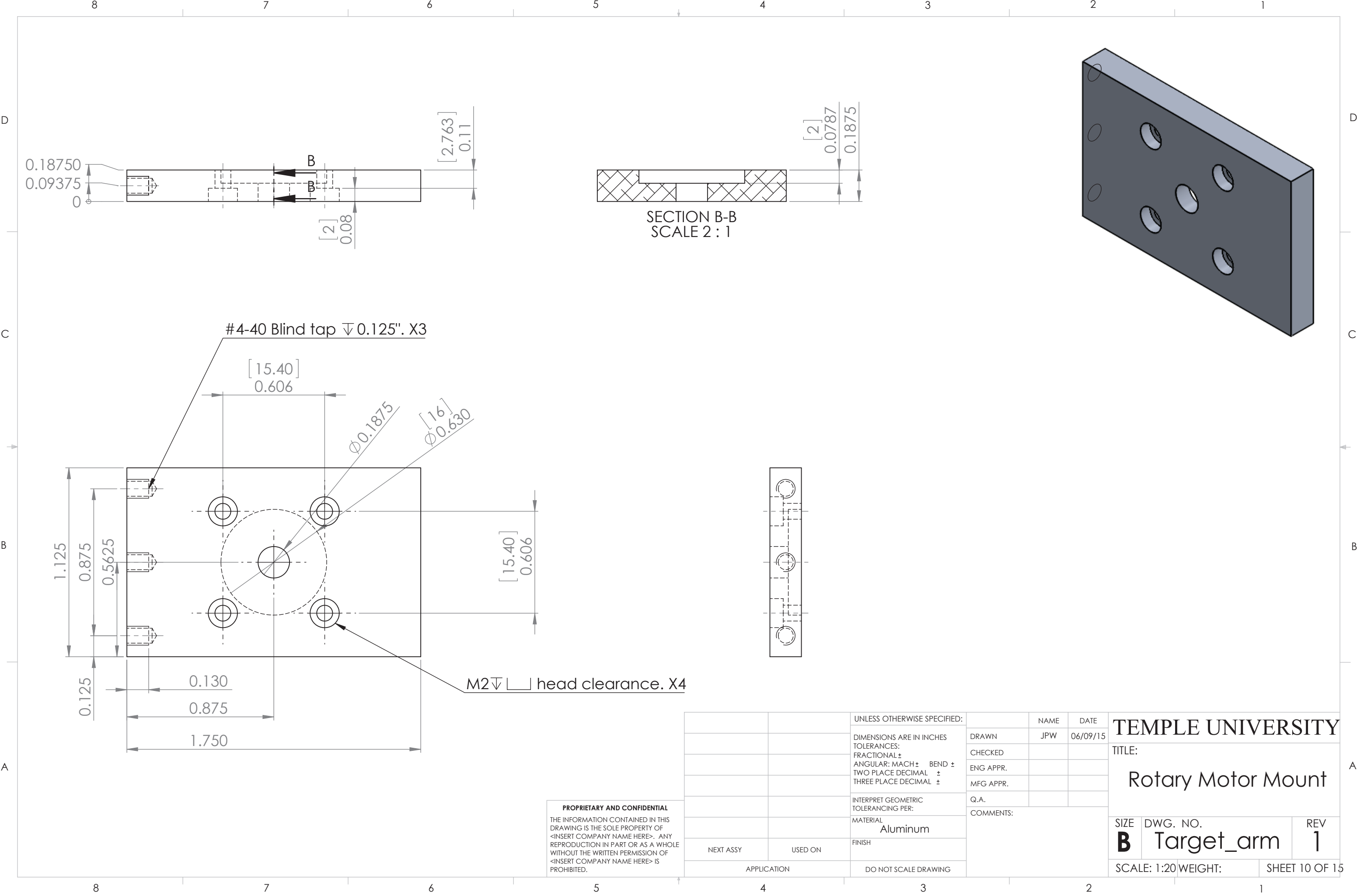
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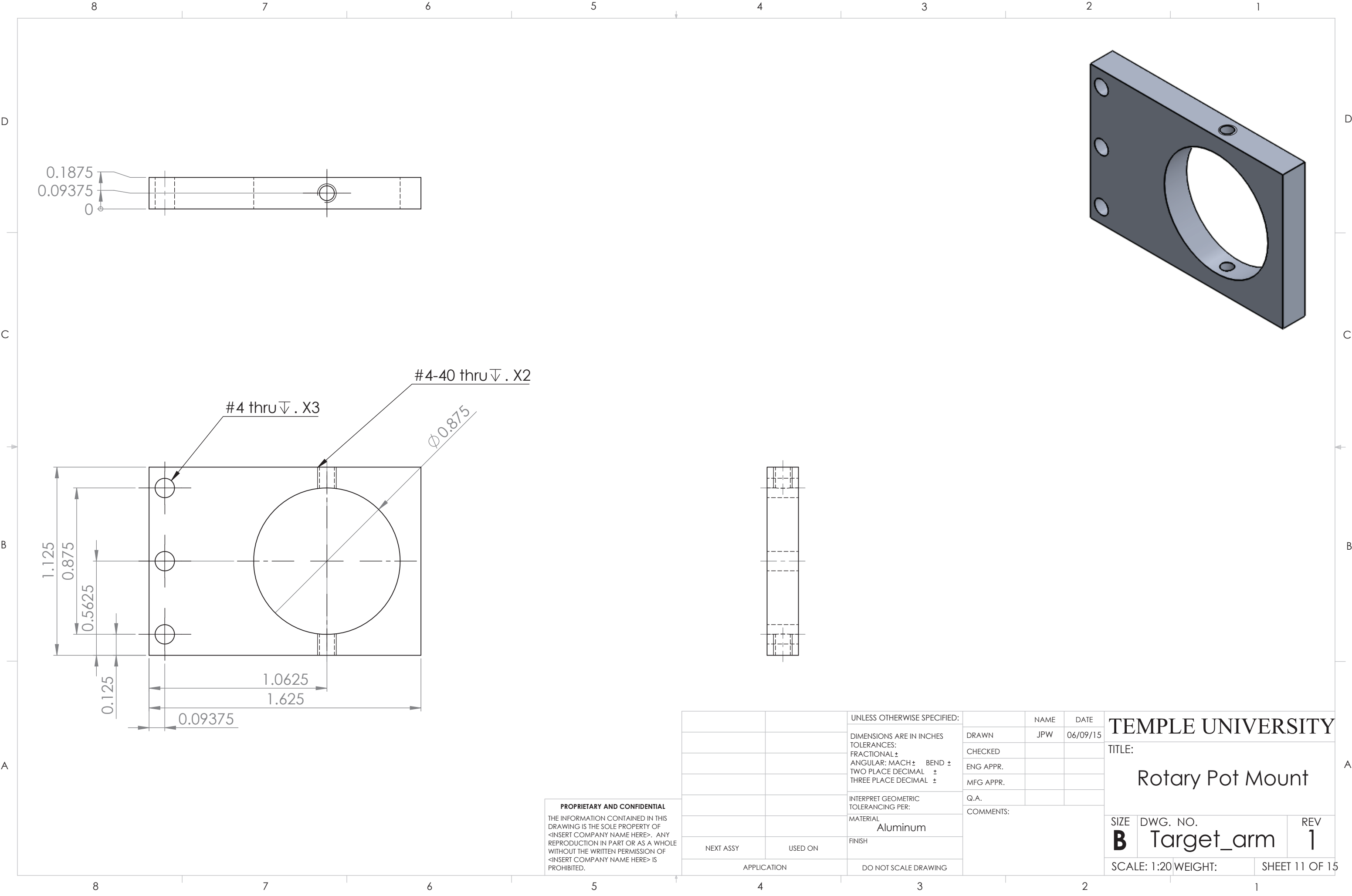
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		DIMENSIONS ARE IN INCHES TOLERANCES: FRACTIONAL $\pm$ ANGULAR: MACH $\pm$ BEND $\pm$ TWO PLACE DECIMAL $\pm$ THREE PLACE DECIMAL $\pm$	DRAWN	JPW	06/09/15		
		INTERPRET GEOMETRIC TOLERANCING PER:	CHECKED				
		MATERIAL Aluminum	ENG APPR.				
		FINISH	MFG APPR.			TITLE: Rotary Mount Bar	
			Q.A.				
			COMMENTS:			SIZE B	DWG. NO. Target_arm
						REV 1	
						SCALE: 1:20 WEIGHT:	
						SHEET 9 OF 15	



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			DRAWN	JPW	06/09/15		
			CHECKED				
			ENG APPR.				
			MFG APPR.				
			Q.A.			SIZE DWG. NO. REV B Target_arm 1	
			COMMENTS:				
NEXT ASSY	USED ON	DO NOT SCALE DRAWING		SCALE: 1:20 WEIGHT:		SHEET 10 OF 15	



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		DIMENSIONS ARE IN INCHES TOLERANCES: FRACTIONAL ± ANGULAR: MACH ± BEND ± TWO PLACE DECIMAL ± THREE PLACE DECIMAL ±	DRAWN	JPW	06/09/15	TITLE: Rotary Pot Mount	
	CHECKED						
	ENG APPR.						
	MFG APPR.						
	Q.A.						
		INTERPRET GEOMETRIC TOLERANCING PER:	COMMENTS:			SIZE DWG. NO. REV B Target_arm 1	
	MATERIAL						
	Aluminum						
NEXT ASSY	USED ON	FINISH					
APPLICATION		DO NOT SCALE DRAWING					
					SCALE: 1:20	WEIGHT:	SHEET 11 OF 15

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1

ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	lever_arm_support	New Part	2
2	lever_arm	New Part	1
3	bearing_shaft	Miniature 316 Stainless Steel Drive Shaft, 1/8" OD	2
4	57155K347	Mini High-Precision Stainless Steel Ball Bearing - ABEC-5, Standard Open, 1/8" ID, 5/16" OD, .109" Width	4

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		UNLESS OTHERWISE SPECIFIED:		NAME	DATE	TEMPLE UNIVERSITY	
		DIMENSIONS ARE IN INCHES	DRAWN	JPW	06/09/15		TITLE:
		TOLERANCES:	CHECKED				
		FRACTIONAL ±	ENG APPR.				
		ANGULAR: MACH ± BEND ±	MFG APPR.				
		TWO PLACE DECIMAL ±	Q.A.			SIZE DWG. NO. REV	
		THREE PLACE DECIMAL ±	COMMENTS:				B Target_arm 1
		INTERPRET GEOMETRIC TOLERANCING PER:					
		MATERIAL				SCALE: 1:20 WEIGHT: SHEET 12 OF 15	
		FINISH					
NEXT ASSY	USED ON	APPLICATION	DO NOT SCALE DRAWING				

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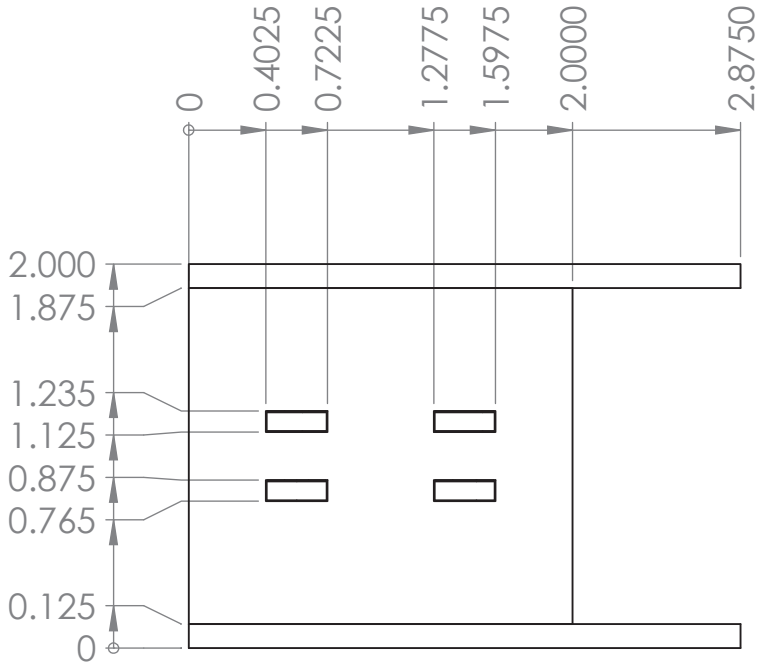
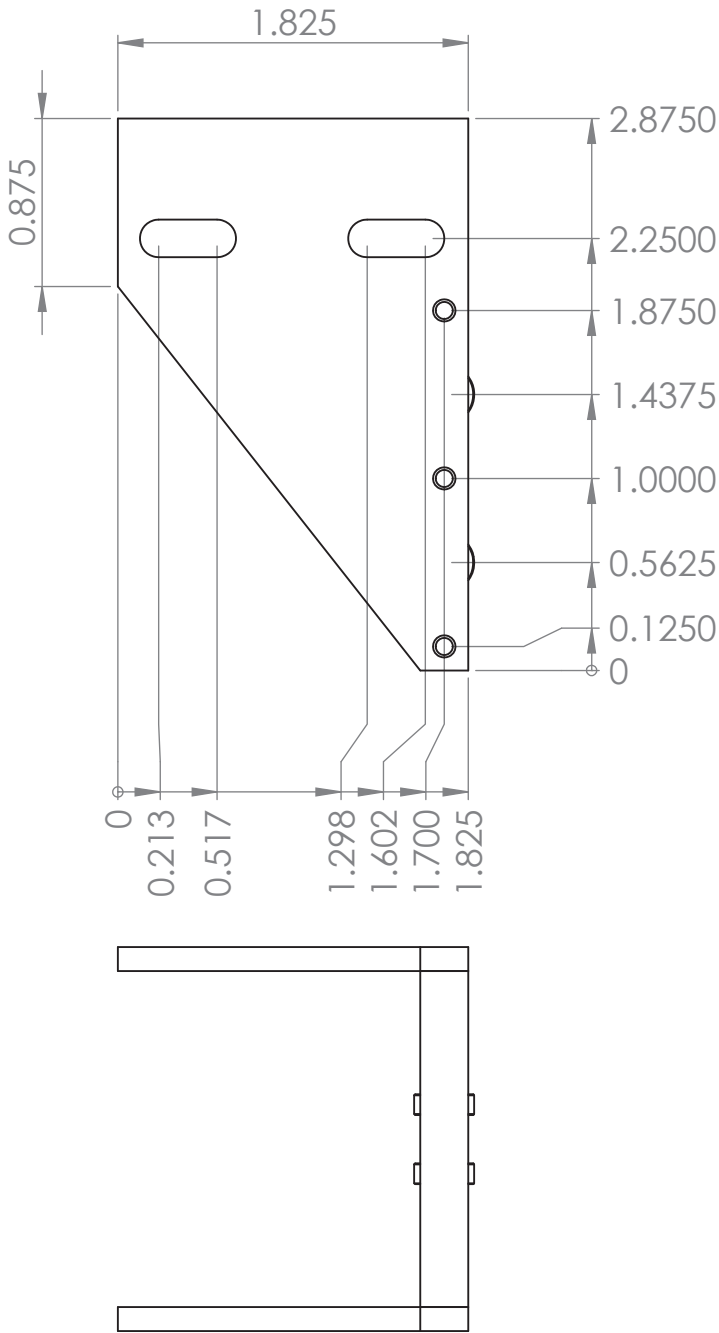
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Part #26 on target-arm assembly



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		DIMENSIONS ARE IN INCHES	DRAWN	JPW	06/09/15		
		TOLERANCES:	CHECKED				
		FRACTIONAL ±	ENG APPR.				
		ANGULAR: MACH ± BEND ±	MFG APPR.			TITLE: Lever_arm	
		TWO PLACE DECIMAL ±	Q.A.				
		THREE PLACE DECIMAL ±	COMMENTS:			SIZE	DWG. NO.
		INTERPRET GEOMETRIC				B Target_arm	
		TOLERANCING PER:					
		MATERIAL				REV	1
NEXT ASSY	USED ON	FINISH				SCALE: 1:20	
APPLICATION		DO NOT SCALE DRAWING				WEIGHT:	SHEET 13 OF 15

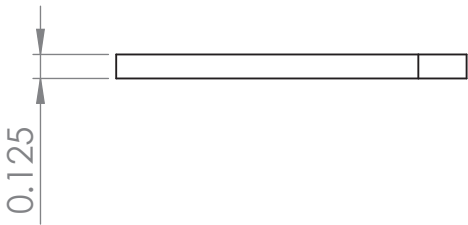
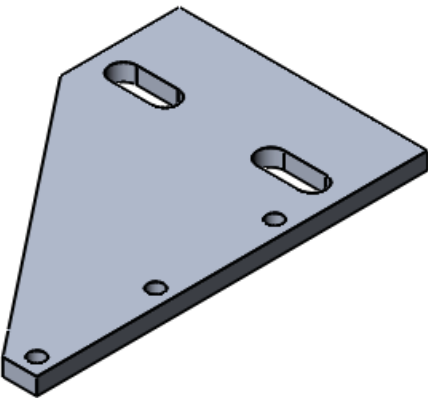
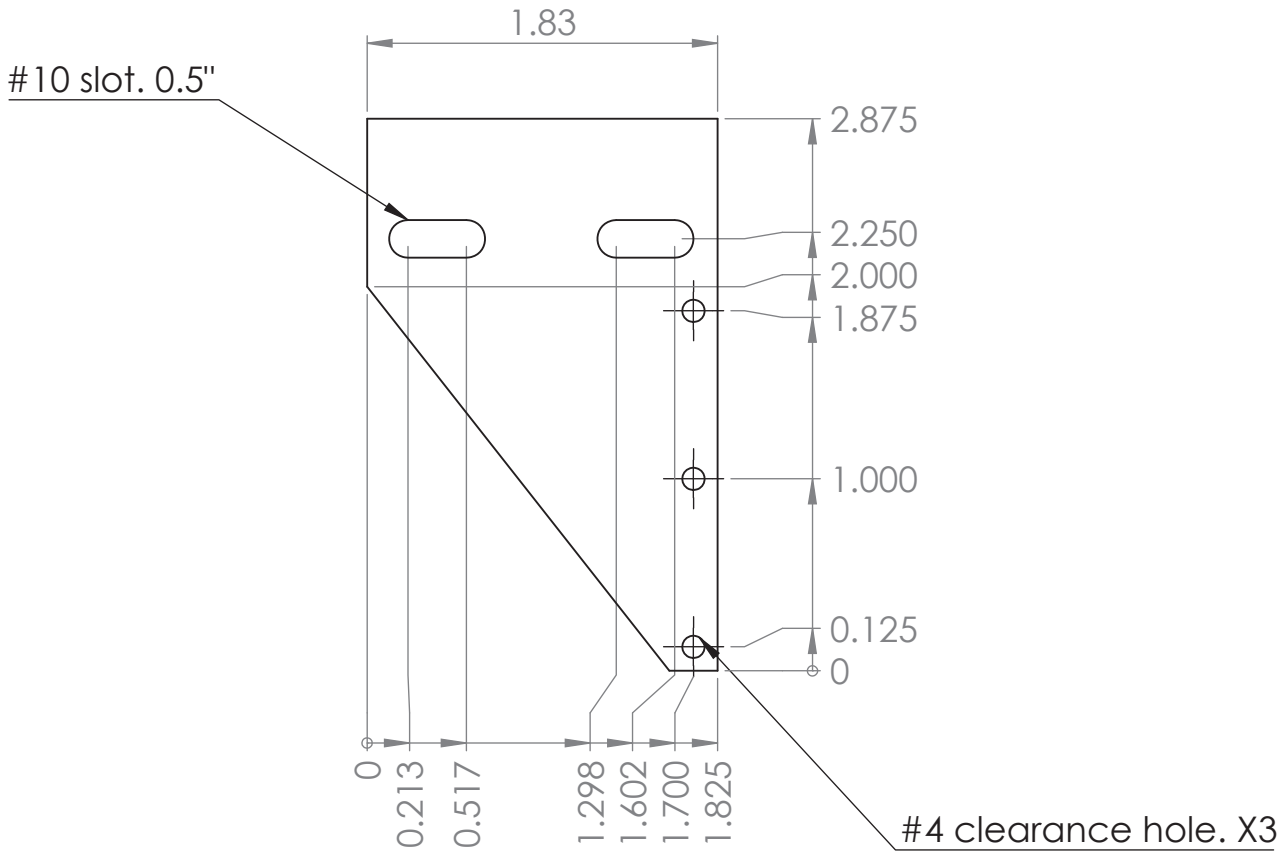
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Part#1 on Lever_arm assembly



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		DIMENSIONS ARE IN INCHES	DRAWN	JPW	06/09/15		
		TOLERANCES:	CHECKED				
		FRACTIONAL ±	ENG APPR.				
		ANGULAR: MACH ± BEND ±	MFG APPR.				
		TWO PLACE DECIMAL ±	Q.A.			TITLE: lever_arm_support	
		THREE PLACE DECIMAL ±	COMMENTS:				
		INTERPRET GEOMETRIC TOLERANCING PER:					
		MATERIAL				SIZE	DWG. NO.
		FINISH				B Target_arm	
NEXT ASSY	USED ON					REV	1
APPLICATION		DO NOT SCALE DRAWING				SCALE: 1:20 WEIGHT:	
						SHEET 14 OF 15	

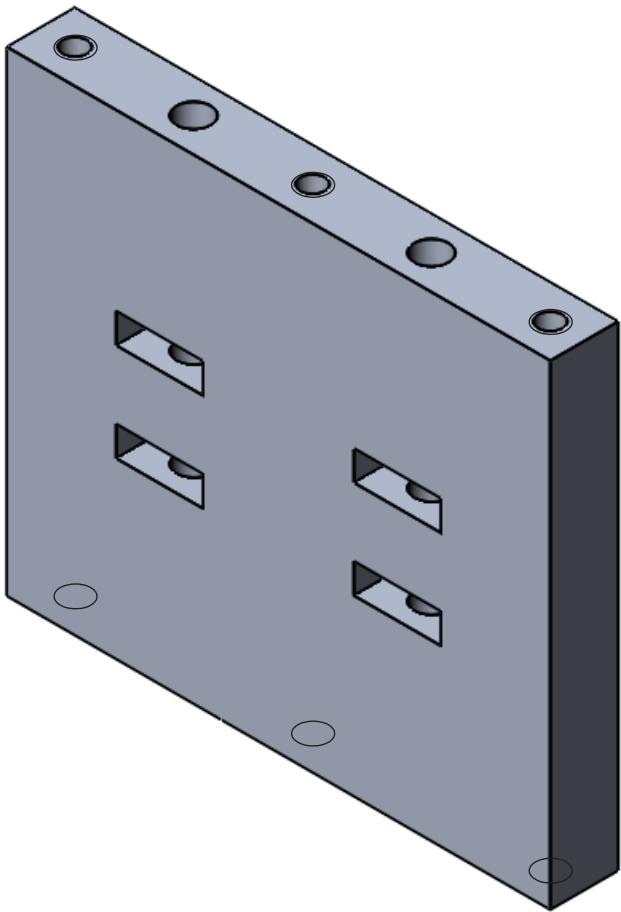
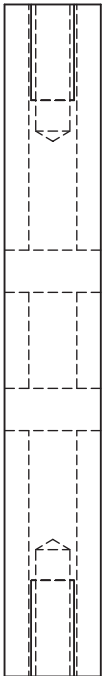
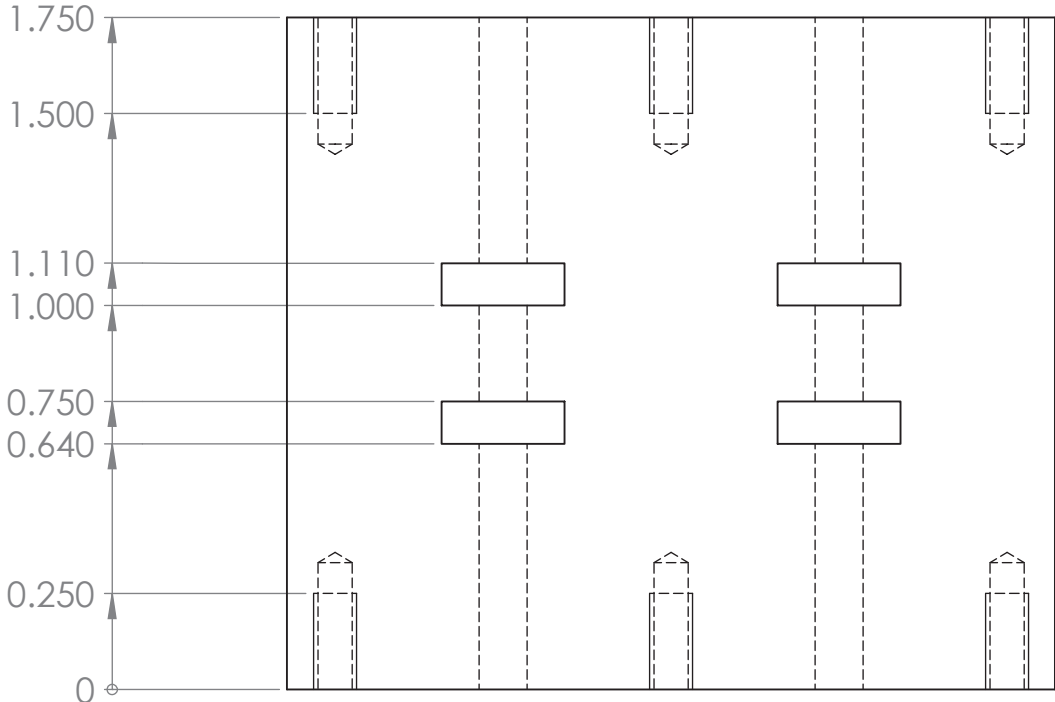
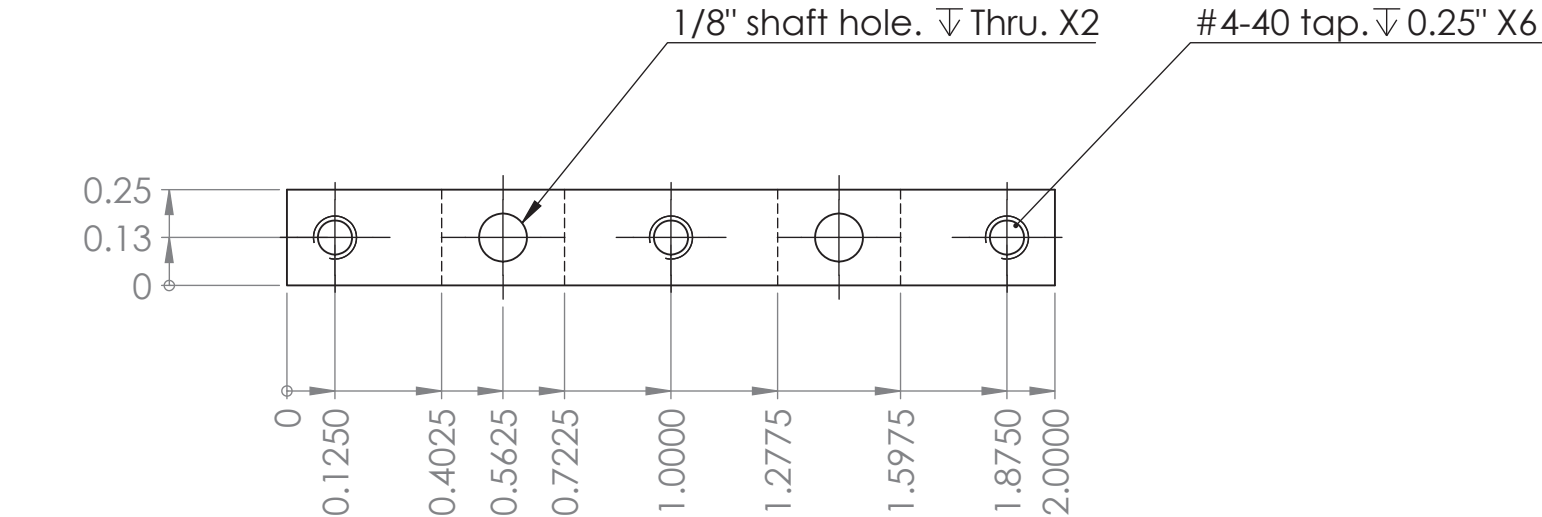
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Part #2 on Lever_arm bill of materials



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		UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN INCHES TOLERANCES: FRACTIONAL ± ANGULAR: MACH ± BEND ± TWO PLACE DECIMAL ± THREE PLACE DECIMAL ±		NAME	DATE	TEMPLE UNIVERSITY			
			DRAWN	JPW	06/09/15				TITLE: Lever_arm part
			CHECKED						
			ENG APPR.						
			MFG APPR.						
		INTERPRET GEOMETRIC TOLERANCING PER:	Q.A.						
		MATERIAL	COMMENTS:			SIZE	DWG. NO.	REV	
NEXT ASSY	USED ON	FINISH				B	Target_arm	1	
APPLICATION		DO NOT SCALE DRAWING				SCALE: 1:20		WEIGHT:	SHEET 15 OF 15